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Carbon sequestration of Chinese forests from 2010 to 2060: spatiotemporal dynamics and its regulatory strategies

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ABSTRACT

Forestation is important for sequestering atmospheric carbon, and it is a cost-effective and nature-based solution (NBS) for mitigating global climate change. Here, under the assumption of forestation in the potential plantable lands, we used the forest carbon sequestration (FCS) model and field survey involving 3365 forest plots to assess the carbon sequestration rate (CSR) of Chinese existing and new forestation forests from 2010 to 2060 under three forestation and three climate scenarios. Without considering the influence of extreme events and human disturbance, the estimated average CSR in Chinese forests was $0.358 \pm 0.016 \text{ Pg C a}^{-1}$, with partitioning to biomass ($0.211 \pm 0.016 \text{ Pg C a}^{-1}$) and soil ($0.147 \pm 0.005 \text{ Pg C a}^{-1}$), respectively. The existing forests account for approximately 93.5% of the CSR, which will peak near 2035, and decreasing trend was present overall after 2035. After 2035, effective tending management is required to maintain the high CSR level, such as selective cutting, thinning, and approximate disturbance. However, new forestation from 2015 in the potential plantable lands would play a minimal role in additional CSR increases. In China, the CSR is generally higher in the Northeast, Southwest, and Central-South, and lower in the Northwest. Considering the potential losses through deforestation and logging, it is realistically estimated that CSR in Chinese forests would remain in the range of $0.161\text{--}0.358 \text{ Pg C a}^{-1}$ from 2010 to 2060. Overall, forests have the potential to offset 14.1% of the national anthropogenic carbon emissions in China over the period of 2010–2060, significantly contributing to the carbon neutrality target of 2060 with the implementation of effective management strategies for existing forests and expansion of forestation.

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1. Introduction

Carbon sequestration (CS) in terrestrial ecosystems, especially forests, has large environmental and economic impacts and provides negative feedback to global warming [1]. Terrestrial ecosystems are estimated to account for approximately 30% of the total anthropogenic carbon dioxide (CO₂) emissions to the atmosphere [2]. In contrast to the oceans, CS in terrestrial ecosystems can be increased or decreased by planting, vegetation removal, and land-use changes, which can be managed more easily [3]. Globally, forests are important natural carbon sinks, and forestation is

considered an effective approach for enhancing terrestrial carbon storage in the context of climate change mitigation [4,5].

Forests hold about 80% of the aboveground biomass C and about 40% of the below-ground C in terrestrial ecosystems [6]. In recent decades, many studies have been conducted to evaluate the carbon storage and sequestration potentials in Chinese forests using model simulations and statistical methods (Table S1 online). Overall, the total carbon storage and its spatial distribution in Chinese forests in the 2010s and the CS rate (CSR) over the past three decades are relatively clear [7–12]. Many studies have evaluated the carbon storage and sequestration potentials in the vegetation of Chinese forests. However, few studies have explored the periodical changes in soil carbon storage due to a lack of primary data at the national scale, and even fewer studies have been conducted on

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predicting the spatiotemporal dynamics of carbon storage changes in the forest ecosystems of China. Based on the State Forestry Administration, forestation in China has been covering a land area of $3.3 \times 10^6 \text{ km}^2$ since 1952, accounting for one-third of the total forested land globally during the same period [13]. Moreover, other ecological projects, such as the Three Northern Protection Forests Project, the Yangtze River Protection Forests Project, and the Pearl River Protection Forests Project, etc., have significantly contributed to the forest carbon stocks in China. Lu et al. [14] reported that the six major ecological restoration projects since the late 1970s have markedly contributed to CS in China from 2001 to 2010, accounting for 56% of the CS in the areas where they were implemented, which is equivalent to 50%–70% of the total annual CS of the terrestrial ecosystems of China over the same period or 9.4% of the carbon emissions from fossil-fuels [15]. Chinese forests are expected to have considerable potential for sequestering atmospheric CO_2 due to their largely immature stand ages [7].

Accurately assessing the CSR of forests at the national scale is essential for developing regional forest carbon strategies, assessing the efforts and benefits of restoring degraded ecosystems, and devising carbon trading mechanisms, etc. These require a proper understanding of the dynamics of the CSR in existing forests with natural restoration. However, regional changes in the CSR and its dynamics could complicate this [7]. Disturbances, such as fire, insect outbreaks, windthrow, and harvesting, markedly affect the carbon cycle in the short term by reducing the leaf area and killing or removing trees, which, in combination with the increased amount of decaying materials, may cause disturbed forest stands to become a temporary CO_2 source [16]. Considering effective measures to enhance the CSR of forests, processes such as selective logging and thinning can ensure an adequate stand density and promote forest growth as well as increase the actual biomass and carbon storage of forests [17]. In addition to maintaining the CSR of existing forests, forestation can enhance carbon fixation, thereby offsetting anthropogenic CO_2 emissions, which is considered the most effective and eco-friendly measure [18–20]. Overall, detailed information regarding the CSR across regions and vegetation types by considering existing forests vs. forestation, vegetation vs. soil, and the turning point of the CSR is essential for achieving carbon neutrality and mitigating future global warming.

Here, we used a forest carbon sequestration (FCS) model [7] based on a classical logistic equation between forest age and biomass to evaluate the dynamics of the CSR for existing forests and three forestation scenarios ($P_{\text{tree}30}$, $P_{\text{tree}40}$, and $P_{\text{tree}50}$) in China under three future climate scenarios (RCP 2.6, 4.5, and 8.5). The main objectives of this study were to identify the (1) dynamics of the CSRs in existing Chinese forests and forestation from 2010 to 2060, (2) relative contributions from existing forests vs. forestation and vegetation vs. soils, and (3) capacity of Chinese forests to offset anthropogenic carbon emissions. Moreover, we aimed to discuss future national forest management strategies that could help to increase the contributions of Chinese forests for achieving carbon neutrality.

2. Materials and methods

2.1. Model for estimating carbon storage in forest vegetation and soils

Based on the classical logistic equation of forest age and biomass, the FCS model was established by He et al. [7] in 2017 to explore the carbon cycle among vegetation, litter, and soil. It was further revised by Yan et al. [21] to estimate the spatiotemporal variations in biomass energy in the terrestrial ecosystems of China from 1980 to 2060. Detailed information regarding the FCS model is provided in the supplementary text files. To consider the impacts of herbivores, fire, water erosion, and forestry activities on carbon

sinks, we used the results of the integrative analysis conducted by Wang et al. [22] and regarded 58% in the ideal state as the result.

2.1.1. Vegetation and soil carbon cycles in the FCS model

With forest development, forest biomass gradually reaches a relative equilibrium state [23]. The relationship between vegetation biomass and forest age can be portrayed using a logistic growth equation [24,25]. The annual growth rate of vegetation biomass can be simply defined by

$$\frac{dB_t}{dt} = v_0 \left(1 - \frac{B_t}{B_{\max}}\right) B_t, \quad (1)$$

where B_t is the forest vegetation biomass (Mg ha^{-1}); V_0 is the intrinsic growth rate, representing the maximum growth rate when vegetative growth is not limited by the environment, nutrients, or disturbances; $B_{\max}$ is the maximum vegetation biomass under the mature forest scenario (Mg ha^{-1}); $1 - B_t/B_{\max}$ represents the fractional deficiency of the current biomass to its saturation level; and t is the forest age (year).

The soil organic carbon (SOC) level in forests is the result of the combined effects of the humification and mineralization processes. When the factors of forests are relatively stable, the humification and mineralization processes will tend to balance the SOC level, as shown in

$$\frac{dC_t}{dt} = I_t - k_2 C_t, \quad (2)$$

where C_t is the SOC density (Mg C ha^{-1}); I_t is the annual input of SOC ($\text{Mg ha}^{-1} \text{ a}^{-1}$); $I_t = h(k_1 L_t)$, L_t is the litter content (Mg ha^{-1}); k_1 is the litter decomposition coefficient; k_2 is the SOC decomposition rate (a^{-1}); and h is the decay coefficient (0.3) [26].

Eq. (2) was then expressed as

$$C_t = \frac{I_t}{k_2} - \left(\frac{I_t}{k_2} - C_{t_0}\right) \exp(-k_2(t - t_0)), \quad (3)$$

where C_t is the soil organic carbon density (Mg C ha^{-1}), I_t is the annual input of soil organic matter ($\text{Mg ha}^{-1} \text{ a}^{-1}$), and k_2 is the rate of SOC decomposition (a^{-1}).

During the development of the FCS model, we used the multi-source investigation data from China over the past few decades to optimize the key parameters, as shown in the [Supplementary materials](#).

2.2. Acquisition of the key parameters

2.2.1. Existing forest data

The main parameters of the FCS model are the initial vegetation biomass (B_0), stand age (t), mean annual temperature (MAT, $^{\circ}\text{C}$), and mean annual precipitation (MAP, mm). The data for the existing forest vegetation biomass were obtained from the field survey of the Strategic Priority Research Program of the Chinese Academy of Sciences (XDA05050000), including the initial vegetation biomass, forest age, litter, and SOC content in 3365 forest sample plots in China, which are the same as the special features of *Proc Natl Acad Sci USA* in 2018 [8,15]. In detail, they included deciduous broadleaf forests (DBF, 806 plots), deciduous coniferous forests (DNF, 197 plots), evergreen broadleaf forests (EBF, 620 plots), evergreen coniferous forests (ENF, 1461 plots), and mixed coniferous and broadleaf forests (NBF, 281 plots; [Fig. S1](#) online) [7]. The areas of different forest types were extracted from the China Remote Sensing Classification system produced by the Strategic Priority Research Program of the Chinese Academy of Sciences (XDA05050000). The areas of DBF, DNF, EBF, ENF, and NBF were 76.27×10^4 , 10.90×10^4 , 35.67×10^4 , 57.86×10^4 , and $9.15 \times 10^4 \text{ km}^2$, respectively [27].

Climate data, including the mean annual temperature (MAT, °C) and mean annual precipitation (MAP, mm), were obtained from the National Climate Center (<http://ncc.cma.gov.cn/cn/>), simulated using the Regional Climate Model system (RegCM 4.0), and then output (spatial resolution of $1^\circ \times 1^\circ$) through one-way nesting of the Beijing Climate Center_Climate System Model Version 1.1 (BCC_CSM1.1) [28].

2.2.2. Forestation data for China

(i). *Forestation area identification.* In theory, all areas suitable for tree growth could be forests. To ensure basic food security, ecological safety, and socio-economic activities, farmland (slopes $<25^\circ$), wetlands, water, tundra, impervious surfaces, and snow and ice are classified as unsuitable for conversion to forests in China. Therefore, current shrubland, grasslands, bare land, and fallow land (slopes $>25^\circ$) are potential areas for forestation.

In this study, we identified the areas of forestation following Ahrends et al. [29], by modeling for global tree cover at $>1\%$, $>25\%$, and $>50\%$ canopy cover combined with two published high-resolution global tree cover datasets [30,31]. With the assistance of environmental modeling, the amount of habitat suitable for tree cover was predicted. Commonly, forests in China are defined as the canopy cover of the trees being greater than 20%, we therefore chose the closest to the crown density as 25% of the trees, in order to identify the suitability of as Ahrends et al. [29] trees grow. To compare and develop suitable forestation schemes and plans, we set up three different scenarios, in which 30% (P_{tree30}), 40% (P_{tree40}), and 50% (P_{tree50}) of the area was classified as suitable for trees. The spatial distribution for suitable forestation under the three scenarios was then determined as the [Supplementary materials](#) (Text S1, Figs. S2 and S3, and Tables S2 and S3 online).

The potential forestation areas of each province were selected according to national planning (“National Major Projects for the Protection and Restoration of Major Ecosystems (2021–2035)”; http://www.gov.cn/zhengce/zhengceku/2020-06/12/content_5518982.htm; <http://www.forestry.gov.cn/main/195/20180108/1064449.html>), and the potential tree plantation areas under each scenario were randomly and equally divided in time and space. The latitude and longitude of each potential tree plantation area were then obtained and the forestation parameters were obtained using the Multi-value Extraction to Point tool in ArcMap 10.2.

(ii). *Key forestation parameters for the FCS model.* In this study, it was assumed that the biomass density of forestation in their first year was 0.1 t C ha^{-1} , and t_0 was the year of planting. The primary soil organic carbon (SOC) content data were obtained from the synthetic field survey dataset [9], and the primary litter data were obtained from the dataset provided by Wen and He [32,33]. The climate data used in these forestation plots were also obtained from these public datasets same as existing forest.

2.3. Supplementary data

Population data for each province were derived from *China Statistical Yearbook 2020* [34] (<http://www.stats.gov.cn/tjsj/ndsj/>), and the CSR per capita was calculated for each province. The CO_2 emission data were obtained from the Open-Data Inventory for Anthropogenic Carbon Dioxide (ODIAC), and the final data were used to calculate the average carbon dioxide emissions from 2010 to 2016 [35,36].

2.4. Statistical analysis

We used a coefficient of 0.5 to transfer biomass to carbon storage [37,38]. We then calculated the changes in carbon density (or CSR) in the vegetation, litter, and different soil layers in five-year increments. The CSRs under the different climate scenarios were treated as replicates, and the data were represented as the average of the three climate scenarios. Furthermore, these data were calculated up to regional or national levels during the analyses.

The spatial distributions of the forests and their carbon densities were produced using ArcMap 10.2 (ESRI, Redlands, CA, USA), and the data were analyzed and plotted on graphs using SPSS 25.0 (IBM Corp., Chicago, IL, USA) and Origin 2018 (Origin Lab, Northampton, MA, USA).

3. Results

3.1. CSRs in Chinese forests from 2010 to 2060

As expected, higher CSRs were observed in Chinese forests from 2010 to 2060, irrespective of the vegetation and soils, existing forests, and forestation level (Fig. 1 and Table 1). The average estimated CSR was $0.358 \pm 0.016 \text{ Pg C a}^{-1}$, with the vegetation and

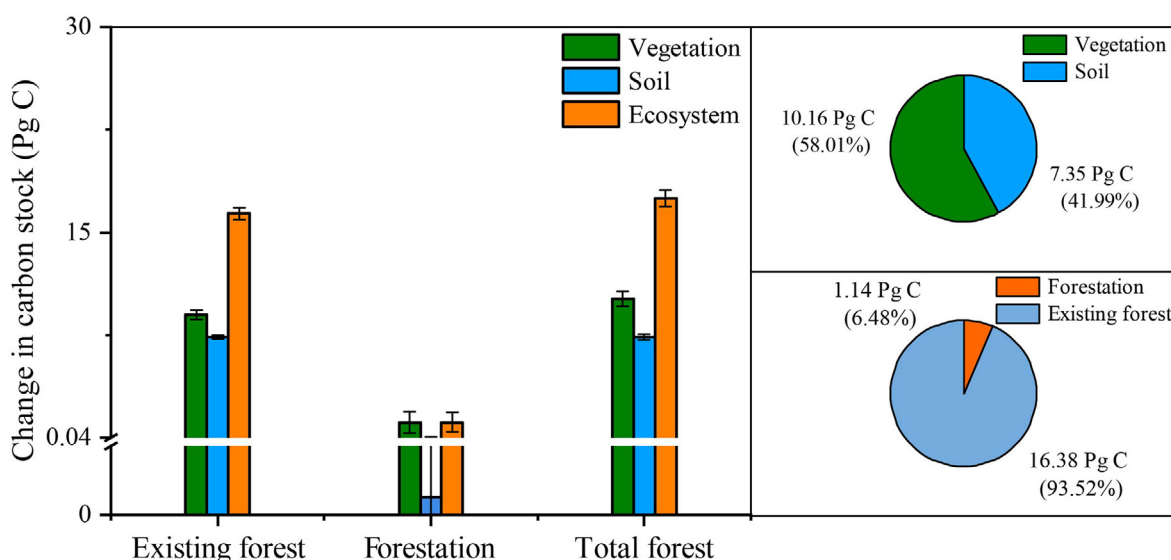


Fig. 1. Carbon sequestration (CS, Pg C) in Chinese forests from 2010 to 2060 and the relative contributions of different components. Bars indicate the averages of different climate and forestation scenarios.

Table 1
Changes in the forest carbon sequestration rate (CSR, Pg C a⁻¹) in China from 2010 to 2060.

Periods	Existing forest (Pg C a ⁻¹)			Forestation (Pg C a ⁻¹)			Total forest (Pg C a ⁻¹)		
	Vegetation	Soil	Ecosystem	Vegetation	Soil	Ecosystem	Vegetation	Soil	Ecosystem
2010–2015	0.177 ± 0.004ab ^{a)}	0.055 ± 0.001a	0.232 ± 0.003ab	–	–	–	0.179 ± 0.004a	0.054 ± 0.001a	0.232 ± 0.003a
2015–2020	0.200 ± 0.007ab	0.105 ± 0.004b	0.305 ± 0.010bc	0.011 ± 0.005a ^{b)}	–0.010 ± 0.006a	0.001 ± 0.001a	0.209 ± 0.007a	0.097 ± 0.004b	0.306 ± 0.010abc
2020–2025	0.232 ± 0.019a	0.141 ± 0.005bc	0.373 ± 0.024c	0.014 ± 0.007b	–0.011 ± 0.006b	0.002 ± 0.001b	0.245 ± 0.019a	0.130 ± 0.005bc	0.375 ± 0.024cd
2025–2030	0.216 ± 0.017ab	0.166 ± 0.007cd	0.382 ± 0.024c	0.018 ± 0.010c	–0.012 ± 0.006b	0.007 ± 0.003c	0.234 ± 0.017a	0.155 ± 0.007cde	0.389 ± 0.024cd
2030–2035	0.208 ± 0.045ab	0.179 ± 0.020cd	0.387 ± 0.065c	0.022 ± 0.015d	–0.009 ± 0.008c	0.013 ± 0.007d	0.230 ± 0.045a	0.170 ± 0.020de	0.400 ± 0.065cd
2035–2040	0.189 ± 0.016ab	0.179 ± 0.008cd	0.369 ± 0.024c	0.028 ± 0.019e	–0.006 ± 0.007d	0.022 ± 0.012e	0.217 ± 0.016a	0.173 ± 0.008de	0.390 ± 0.024cd
2040–2045	0.194 ± 0.022ab	0.193 ± 0.015d	0.387 ± 0.036c	0.030 ± 0.022f	0.001 ± 0.007e	0.032 ± 0.019f	0.225 ± 0.022a	0.194 ± 0.015e	0.419 ± 0.037d
2045–2050	0.147 ± 0.058b	0.162 ± 0.042cd	0.309 ± 0.100bc	0.035 ± 0.027g	0.007 ± 0.007f	0.042 ± 0.027g	0.182 ± 0.058a	0.169 ± 0.043de	0.351 ± 0.100bcd
2050–2055	0.166 ± 0.075ab	0.174 ± 0.038cd	0.340 ± 0.113c	0.035 ± 0.025g	0.016 ± 0.010g	0.051 ± 0.034h	0.201 ± 0.076a	0.190 ± 0.038e	0.391 ± 0.114cd
2055–2060	0.075 ± 0.020c	0.118 ± 0.018b	0.193 ± 0.037a	0.037 ± 0.027h	0.022 ± 0.014h	0.058 ± 0.041i	0.111 ± 0.021b	0.140 ± 0.018cd	0.251 ± 0.037ab
Average	0.180 ± 0.007	0.147 ± 0.003	0.328 ± 0.008	0.025 ± 0.017	0.000 ± 0.002	0.025 ± 0.016	0.211 ± 0.016	0.147 ± 0.005	0.358 ± 0.016

a) Changes in the carbon sequestration rate are presented as the mean ± 1 standard deviation under different climate scenarios (RCP 2.6, 4.5, and 8.5), and the same small letters indicate no significant difference at $P = 0.05$ level.
b) The data of afforestation were the average of three forestation scenarios (30%, 40%, and 50%) under three climate scenarios.

soil contributing to 0.211 ± 0.016 (59%) and 0.147 ± 0.005 Pg C a⁻¹ (41%), respectively. Furthermore, the CSR of the existing forests was approximately 0.328 ± 0.008 Pg C a⁻¹, while that of the forestation since 2010 was 0.025 ± 0.016 Pg C a⁻¹. Therefore, the existing forests played a more predominant role in sequestering atmospheric CO₂ (93.5%) than new planted forests (6.5%).

3.2. CSR dynamics in Chinese forests

The CSRs of Chinese forests exhibited clear dynamics from 2010 to 2060. The total CSR, i.e., the sum for all forests, increased from 0.232 ± 0.003 Pg C a⁻¹ in 2010 to 0.419 ± 0.037 Pg C a⁻¹ in 2045 and then decreased to 0.251 ± 0.037 Pg C a⁻¹ in 2060, as expected (Fig. 2, and Tables 1 and S4–S7 online). For the existing forests, the CSR ranged from 0.232 ± 0.003 to 0.387 ± 0.065 Pg C a⁻¹, and peaks from vegetation and soil occurred in 2025 and 2035. Regarding the forestations, the CSR increased from 0.001 ± 0.001 to 0.058 ± 0.041 Pg C a⁻¹ in 2060. Notably, the SOC density in the forestations may have been decreased to some extent at an earlier stage. CSR had a lower effect on soil than vegetation irrespective of whether in existing forests or forestations.

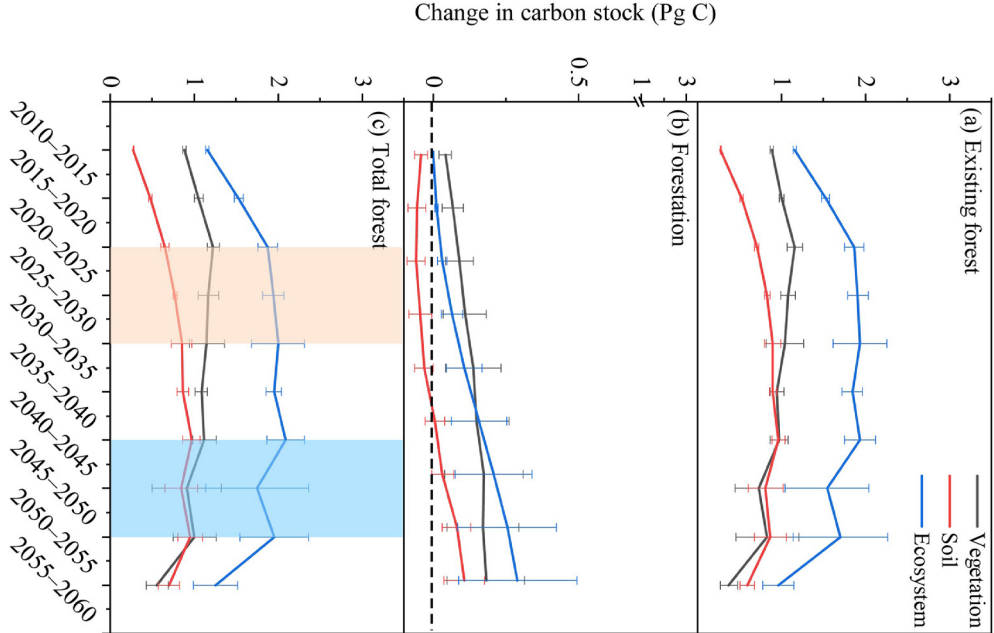


Fig. 2. Carbon sequestration (CS) dynamics (Pg C) in Chinese forests from 2010 to 2060. Bars indicate the averages of different climate and forestation scenarios. (a) and (b) show the change in CS of existing forests and forestation, respectively. As predicted, the peak CS in forests will arrive in 2035, and the CS will most likely decrease rapidly after 2050 without rational human management (c).

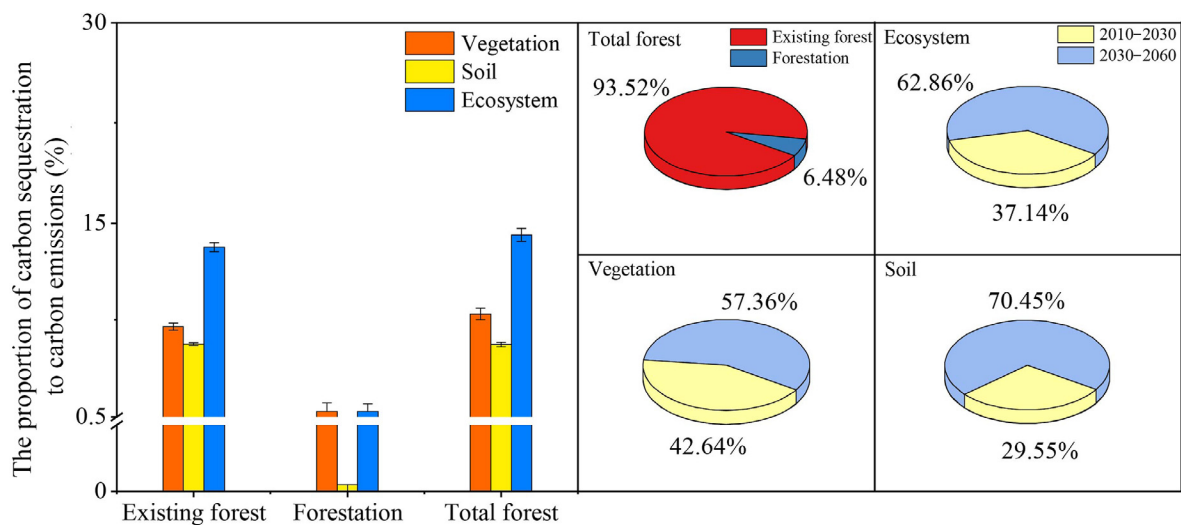
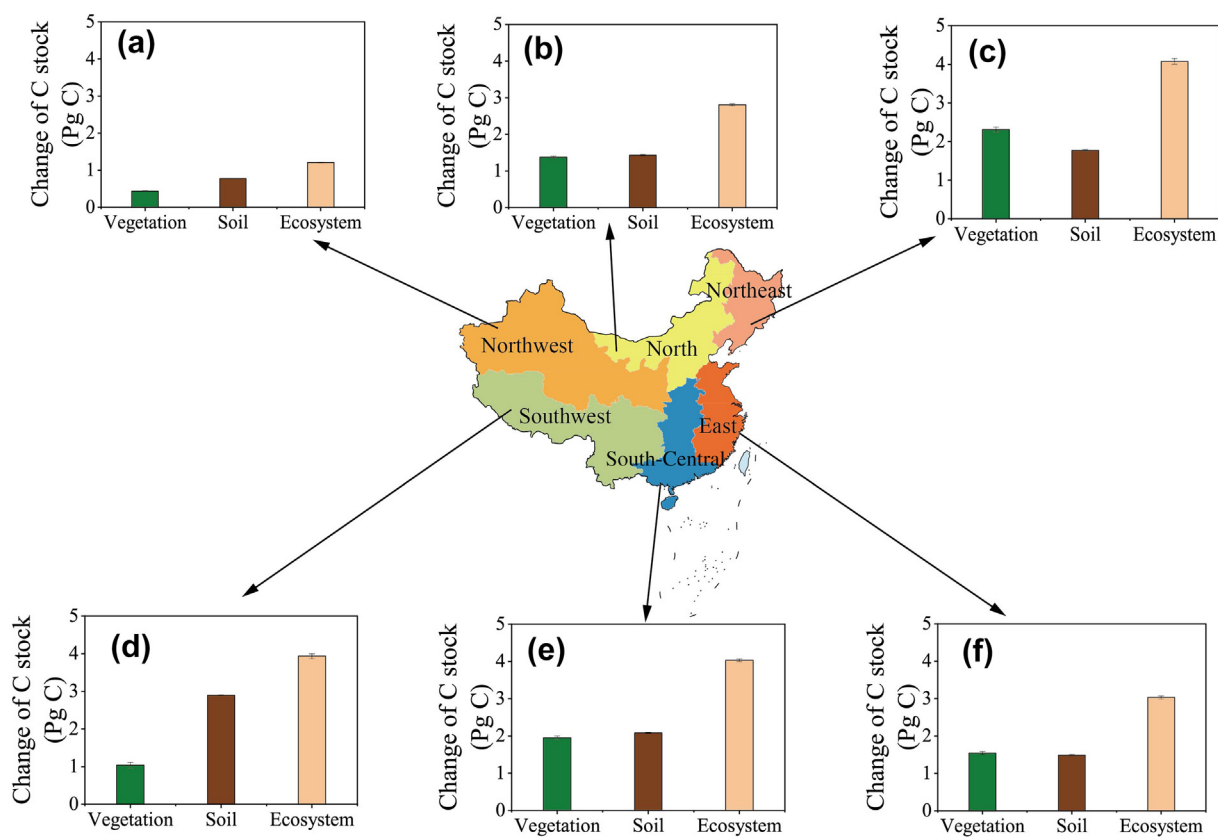


Fig. 3. Capacity for carbon sequestration in forests to offset CO₂ emissions from 2010 to 2060 in China. Bars indicate the averages of different climate and forestation scenarios. The CO₂ emission data were released by the Open-Data Inventory for Anthropogenic Carbon Dioxide (ODIAC, 2016), and the average carbon dioxide emissions from 2010 to 2016 were used for calculation [35].



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Fig. 4. Changes in carbon sequestration (CS, Pg C) in Chinese forests from 2010 to 2060 among the different regions. Data were the averages of the errors of different climatic and forestation scenarios. Forests in China are divided into six regions: (a) Northwest, including Gansu, Ningxia, Qinghai, Shaanxi, and Xinjiang; (b) North, including Hebei, Inner Mongolia and Shanxi, and the cities of Beijing and Tianjin; (c) Northeast, including Heilongjiang, Jilin, and Liaoning; (d) Southwest, including Guizhou, Sichuan, Yunnan and Tibet, and the city of Chongqing; (e) South-Central, including Guangdong, Guangxi, Hainan, Henan, Hubei and Hunan; and (f) East, including Anhui, Fujian, Jiangsu, Jiangxi, Shandong and Zhejiang, and the city of Shanghai. Data for Hong Kong, Macao, and Taiwan were not used in this investigation.

3.3. Spatial variations in CSR

Chinese forests could offset 14.1% of the anthropogenic CO₂ emissions between 2010 and 2060, 13.2% and 0.9% of which could be offset by existing forests and forestations, respectively (Fig. 3). The CSRs of Chinese forests were higher in the Northeast, Southwest, and South-Central regions, but lower in the Northwest (Fig. 4). The relative contributions in these regions varied due to CSRs changing differently. Furthermore, the CSRs were higher in the provinces in the southwestern, southern-central, and north-eastern areas and lower in Xinjiang, Gansu, and Qinghai (Figs. S4 and S5 online). In all provinces, the changes in the CSRs in vegetation were higher than those in soils (Figs. S6 and S7 online).

4. Discussion

4.1. Model uncertainty and limitations

The model was constructed using long-term datasets, which can suitably fit the relationship between vegetation biomass and forest age, but more long-term forest succession datasets are required to reduce the errors, considering the complexity of forest types and their wide distribution in China. At the establishment of the FCS model used here, some key parameters of the model were calibrated according to the investigated data of more than 3000 forest plots and the details seeing our previous research [7], therefore, these representative local parameters made it better simulate the spatial C storage in Chinese forests to a large extent. Furthermore, the dynamic datasets of forest succession (including vegetation biomass, 0–1 m soil carbon density and forest age) were obtained from public databases and were used to validate the applicability of the FCS model in different forests across China. During validation, simulated and measured vegetation biomass (soil carbon density) was close to the 1:1 line, demonstrating that the FCS model can be used to predict forest carbon sequestration capacity well (Fig. S8 online). Additionally, owing to the lack of indigenous parameters, the soil carbon pool only considered the input of litter and did not consider the inputs of microorganisms and other carbon sources. This issue may have led to uncertainty in the estimation of soil carbon storage. Temperature and precipitation were important parameters in the FCS model; however, the estimates might contain high levels of uncertainty under different global climate change scenarios, which can lead to uncertainty of forest carbon storage estimates. In future, we will further improve the FCS model, especially through incorporating into soil microbe, to decrease its uncertainty, because soil C storage is tremendous and has important roles in estimating forest carbon sequestration.

4.2. Chinese forests could potentially sequester large amounts of atmospheric CO₂

The total increase in carbon storage in the forest ecosystems of China from 2010 to 2060 was approximately 17.510 ± 0.608 Pg C, with an average CSR of 0.358 ± 0.016 Pg C a⁻¹. This could offset $14.1\% \pm 0.49\%$ of the anthropogenic CO₂ emissions identified in the emission standard of China for 2015 [36]. Furthermore, the scenarios with 7798 and 537 km² of deforestation in 2010 and 2060 (Text S2 and Table S9 online), respectively, could result in a loss of 0.030 Pg C a⁻¹ from the forest ecosystem of China. Annual logging of 3.34×10^8 m³, as was conducted in the 2010s (Text S2 online), would result in a loss of 0.167 Pg C a⁻¹. Consequently, our CSR estimates are likely to be overestimated but may be achievable if the government reduces the wood requirements for deforestation and logging. For existing forests, the biomass carbon density (50.74 Mg C ha⁻¹) was closer to the results of Tang et al. [8]

for the same period, and these were higher than the results reported in most previous studies (Table S1 online). The differences among the various studies were due to the methods employed, such as different model parameters and differences in sample site selection and data sources, particularly for different periods (changes in forest area and age structure). Some studies have demonstrated that changes in both forest area expansion and growth can accelerate the increase in carbon storage in the vegetation of Chinese forests [39,40]. As shown in Table S1 (online), few studies have explored the periodical changes in soil carbon storage owing to a deficiency of country-wide primary data [11,41]. The carbon density in both the vegetation and soils of the existing forests increased from 2010 to 2060, resulting in an annual CSR of approximately 0.328 ± 0.008 Pg C a⁻¹. The spatial distribution of the CSR indicated that the forests in the northeastern and southwestern regions had stronger capacities to sequester atmospheric CO₂, which is consistent with the results of Wang et al. [36], who used satellite data. Although our estimates were below 0.8 Pg C a⁻¹, they can provide a strong foundation for a strategy by which China could achieve carbon neutrality by 2060, after revising the results.

However, the CSR was probably overestimated in this study due to the uncertain effects of drought-induced tree mortality, human activity, natural disasters, insect infestations, fires, or changes in the areas of the existing forests. The actual area of forestation and future species selection also created uncertainty, which were influenced by various factors, including population growth, demand for agricultural land, and other policies [42]. Different carbon emission data sources also created uncertainty regarding the contributions of forest ecosystems to the carbon balance in China. The predicted increase in forestation in China between 2010 and 2050 is approximately 300,000 ha, which may yield a net carbon sequestration potential of 2.51 Pg C from forestation by 2050 [19], which exceeds our estimates (0.588 ± 0.347 Pg C). One plausible reason for this difference is that the FCS model used three climate scenarios in CMIP5 to consider the impacts of future climate change on CSR.

4.3. The roles of newly planted forests are unexpectedly lower during 2010–2060

The CSR of the new forestations continuously increased from 2010 to 2060, however, their contribution only accounted for 6.5%, while existing forests accounted for 93.5%. After the early period of forestation, vegetation carbon storage usually increases slowly and then accelerates according to a logistic equation [7]. However, in some regions, soil carbon storage could be neutral or even decrease during the early periods (Fig. 2). Therefore, the CSR processes are asynchronous for vegetation and soil, and it is difficult to predict the future CSR dynamics accurately based on the existing forests and new forestation. However, such information can provide a theoretical foundation for regulating the CSR. Forestation activities may destroy the internal structures of soils and alter their geological permeability, which benefits microbes that decompose SOC and can even accelerate soil erosion by wind and water [43]. Gradually, with the increase in new organic matter inputs and restoration of soil structures, soil carbon storage increased to some extent, although it lagged for several years compared with vegetation (Fig. 2).

Although the enthusiasm for forestation in China is high, it currently plays a minimal role in the plans to achieve carbon neutrality by 2060. With the progress of forestation in recent decades, the amount of land suitable for forest growth is gradually decreasing, and the remaining areas for forestation are located in environments with poor soil quality, topography, precipitation, and temperature. Therefore, there will be high risk and low CSR potential

with forestation in these areas. However, the selection of suitable tree species, even if they are mixed forests to prevent the threats of insect outbreaks and diseases, could be promising [16,44].

4.4. High CSR in existing forests requires national management

Existing forests were the predominant sources for the CSR when compared with the new forestations. Based on the measured plot data, the estimated age of Chinese forests was 33.97 years in 2010 (Table S10 online), located in the phase developing into peak biomass in the logistic equation. This indicates that the existing forests had high CSRs (Table 1 and Fig. 2). Based on this, we found that the maximum CSR of the existing forests approximately appeared in the period from 2030 to 2035, and the forest stand age was approximately 40–50 years among the different regions (Tables 1 and S10 online), and this was higher in cold-temperate coniferous forests than in tropical forests [45]. Two important points for the existing forests should be emphasized: (1) their vegetation CSR will begin to decrease in 2035, and (2) the total CSR will begin to decrease in 2050. Regarding the stand ages, we need to take more accurate measures to determine whether they should be modified to allow the existing forests to maintain their high CSRs in the long term. Certain management measures could be implemented to enhance the capacity of Chinese forests and sequester atmospheric CO₂. Such measures include fertilization, fire control, pest control, forest regeneration, young forest tending (including weeding and soil loosening), pruning, thinning, fertilization, irrigation, drainage, harvesting mode selection, rotation, and management of harvest residues and deadwood, can enhance the capacity of forests to further sequester atmospheric CO₂ in China [46–48]. Therefore, the high CSRs in the existing forests will be maintained if effective measures and stand age information can be rationally used on a regional basis for factors such as plant species, climate, and soil.

5. Conclusions

The estimated average carbon sequestration rate (CSR) from 2010 to 2060 was $0.358 \pm 0.016 \text{ Pg C a}^{-1}$, which could offset $14.1\% \pm 0.49\%$ of the anthropogenic CO₂ emissions identified in the 2015 emission standard of China. Although the CSR of the new forestations may increase continuously from 2010 to 2060, their contribution was only 6.5%, with existing forests playing the predominant role. Theoretically, the estimates should be overestimated to some extent, because some extreme events and human disturbance have not been considered here. Two notable points regarding the CSR of existing forests were identified: (1) the vegetation CSR will begin to decrease in 2035, and (2) the total CSR will begin to decrease in 2050. Therefore, if we can implement effective measures to regulate the dynamics of the CSR with various methods, such as selective cutting, thinning, adjusting rotation time, using different stand age combinations, and considering the regional specificity of plant species, climate, or soils, high long-term CSRs may be achieved. The deforestation and logging potentials were used to estimate the CSR, which ranged from 0.161 to $0.358 \text{ Pg C a}^{-1}$. In summary, Chinese forests could potentially offset a large amount of anthropogenic carbon emissions from 2010 to 2060; however, achieving this depends on the appropriate management of the existing forests in the future as same as laying emphasis on forestation in national policies.

Conflict of interest

These authors declare that they have no conflict of interest.

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Author contributions

Weixiang Cai wrote the manuscript, calculated and analyzed the data, and drew the figures. Mingxu Li, Li Xu, Pu Yan, Jianhua Zhu, and Xiaoquan Zhang made significant contributions to the design of the study and revised the manuscript. Longzhu Wang, Xiaoyu Cen, and Guoxin Si were responsible for the options of data-processing and statistical methods. Osbert Jianxin Sun, Nianpeng He, and Guirui Yu conceived the ideas and designed methodology. All authors contributed critically to the drafts and gave final approval for publication.

Appendix A. Supplementary materials

Supplementary materials to this article can be found online at <https://doi.org/10.1016/j.scib.2021.12.012>.

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